Symbolic rules and how a conclusion is reached

Domain-agnostic federated neuro-symbolic framework

Healthcare disease risk and banking fraud

This note lists the symbolic rules the framework actually runs. They are fixed Python checks, not text generated by the neural model. For every case the system adds the points of the rules that fire, maps that total to a class, and only then decides whether the symbolic class replaces the neural class. The written explanation is allowed to mention only the rules that fired.

1. From features to a conclusion

- The saved federated neural model proposes a class and a confidence. That proposal is not the explanation.
- Every symbolic rule is tested against the case features. A rule that does not meet its condition adds 0. A rule that fires adds its fixed points. Negative points lower the score.
- The sum of the fired points is the symbolic score. Thresholds turn that score into a symbolic class.
- If the symbolic class and the neural class are the same, they agree. The conclusion is that class.
- If they differ, the neural class is kept unless an override condition below is met. An override replaces the neural class with the symbolic class. A disagreement that is not overridden is stored as a conflict, and the explanation must say so.
- The explanation is built only from the entity id, the final class, the fired rule names, and the evidence sentences those rules produced.

2. Healthcare rules

Task: classify a patient as low_risk, medium_risk, or high_risk. Ten rules. Protective rules carry negative points.

Rule	Fires when	Points	Evidence recorded
hypertension	Systolic BP >= 140 or diastolic BP >= 90	+1.5	Blood pressure in mmHg
hyperglycemia	Glucose >= 126 mg/dL	+1.4	Glucose value
obesity	BMI >= 30	+1.0	BMI value
smoker	smoker is true	+1.2	Current smoker status is positive
family_history	family_history is true	+1.0	Positive family history
hypoxia	Oxygen saturation < 94%	+1.6	Oxygen saturation
sedentary	Exercise < 1.5 hours/week	+0.6	Weekly exercise hours
advanced_age	Age >= 60	+0.8	Age
protective_exercise	Exercise >= 5 hours/week	-0.7	Protective exercise level
protective_sleep	Sleep is from 7 to 9 hours	-0.4	Sleep duration

3. Healthcare score to class

Symbolic score	Symbolic class
score < 2.0	low_risk
2.0 <= score < 4.0	medium_risk
score >= 4.0	high_risk

Firing one risk rule does not by itself make the class medium or high. The class is the total. A single hypertension rule is +1.5, which is still low_risk.

4. When healthcare replaces the neural class

The neural class is replaced only in these situations. Otherwise it stands.

- Hard constraint, and the neural class is low_risk. This applies when oxygen saturation < 92, or systolic BP >= 180, or glucose >= 250, or the score is at least 4.0 and at least four rules fired. Oxygen below 92, or a score of at least 4.0 with four or more rules, forces high_risk. The other critical readings force medium_risk.
- Medium floor, and the neural class is low_risk: score >= 3.0 and at least four rules fired. The conclusion becomes medium_risk.
- Any disagreement where the neural confidence is below 0.55. The conclusion becomes the symbolic class. This is the override used for patient P00002.
- Any other disagreement: the neural class is kept, and the conflict is written into the explanation.

5. Worked healthcare conclusions

Both patients are from the held-out healthcare test set. The neural class is the saved FedAdagrad model. The symbolic score below was produced by running HEALTHCARE_RULES on these features.

Patient P00000. Neural class: low_risk (confidence 0.972). Features used by the rules: age 52, BMI 16.0, BP 144.6/74.1, glucose 117.1, oxygen 97.2%, smoker no, family history no, exercise 4.31 h/week, sleep 6.69 h.

Fired rule	Why it fired	Points
hypertension	Hypertension indicators: BP 144.6/74.1 mmHg	+1.5
Total symbolic score	score < 2.0, so the symbolic class is low_risk	+1.5

Conclusion: symbolic class low_risk and neural class low_risk. They agree, so the final class is low_risk. The explanation may cite only hypertension. No other healthcare rule fired, so none of those names can appear as applied rules.

Patient P00002. Neural class: low_risk (confidence 0.516). Features: age 58, BMI 21.49, BP 144.2/80.0, glucose 109.8, oxygen 96.3%, smoker yes, family history no, exercise 3.29 h/week, sleep 6.32 h. The test label is medium_risk.

Fired rule	Why it fired	Points
hypertension	Hypertension indicators: BP 144.2/80.0 mmHg	+1.5
smoker	Current smoker status is positive	+1.2
Total symbolic score	2.0 <= 2.7 < 4.0, so the symbolic class is medium_risk	+2.7

Conclusion: the classes differ, and the neural confidence 0.516 is below 0.55, so the override rule applies.

The neural suggestion low_risk is replaced. The final class is medium_risk. The explanation cites hypertension and smoker, states that the neural suggestion was overridden, and does not name any rule that scored 0.

6. Banking rules

Task: classify a transaction as legitimate or fraudulent. Ten rules. There are no negative-point rules in this domain.

Rule	Fires when	Points	Evidence recorded
high_amount	Amount >= 900	+1.2	Transaction amount
amount_spike	Amount / 30-day average >= 3.5	+1.5	Spike ratio
far_from_home	Distance from home >= 80 km	+1.2	Distance in km
geo_jump	Distance from last transaction >= 60 km	+1.0	Geo jump in km
txn_burst	Transactions in 1 hour >= 4	+1.3	Burst count
international	is_international is true	+1.0	International transaction flag
night_txn	is_night is true (hours 00:00-05:59)	+0.8	Night-time transaction
failed_pin	Failed PIN attempts >= 2	+1.6	PIN attempt count
high_risk_merchant	Merchant risk >= 0.65	+1.1	Merchant risk score
card_not_present	card_present is false	+0.7	Card-not-present transaction

7. Banking score to class

Symbolic score	Symbolic class
score < 3.0	legitimate
score >= 3.0	fraudulent

The neural class is replaced only in these situations. Otherwise it stands.

- Clear fraud pattern: score >= 4.5, the neural class is legitimate, and the neural confidence is below 0.75. The conclusion becomes fraudulent.
- PIN and amount together: failed PIN attempts >= 3, amount-versus-average ratio >= 4.0, and the neural class is legitimate. The conclusion becomes fraudulent.
- Any disagreement where the neural confidence is below 0.55. The conclusion becomes the symbolic class.
- Any other disagreement: the neural class is kept, and the conflict is written into the explanation.

8. Worked banking conclusion

Transaction T000000 is the first held-out banking test case. The neural class is the saved FedAvg model: fraudulent, confidence 0.999. The test label is fraudulent. The symbolic score was produced by running BANKING_RULES on these features: amount 145.98, 204.96 km from home, 3.67 km from the last transaction, 4 transactions in one hour, international yes, night no, card present yes, merchant risk 0.01, PIN failures 0.

Fired rule	Why it fired	Points
far_from_home	Far from home location: 205.0 km	+1.2
txn_burst	Transaction burst in 1 hour: count=4	+1.3
international	International transaction flag	+1.0
Total symbolic score	score >= 3.0, so the symbolic class is fraudulent	+3.5

Conclusion: symbolic class fraudulent and neural class fraudulent. They agree, so the final class is fraudulent. No override is required. The explanation cites far_from_home, txn_burst, and international. high_amount, amount_spike, geo_jump, night_txn, failed_pin, high_risk_merchant, and card_not_present did not fire, so they are not part of the conclusion.

9. What this shows

- The class from the symbolic layer is a sum of named rules with published thresholds, not a second neural guess.
- A fired rule is evidence. It becomes the class only when the total crosses 2.0 or 4.0 in healthcare, or 3.0 in banking.
- The neural class is overturned only by the override conditions in sections 4 and 7. P00002 is an overturn. P00000 and T000000 are agreements.
- The explanation cannot introduce a rule that scored zero. That is the check a teacher can trace from the tables above to the sentence the system prints.